

Hulled Rice or husk? Synchrotron radiation XRF and deep learning approach for the determination of the geographical origin of Chinese rice samples

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Highlights

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Synchrotron XRF–deep learning framework developed for rice geographical authentication.

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903 hulled rice and 824 husk samples from 16 provinces were analyzed by SR-XRF.

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2D-AlexNet analyses prove higher traceability accuracy of husk (99.07%) vs. grain (98.02%).

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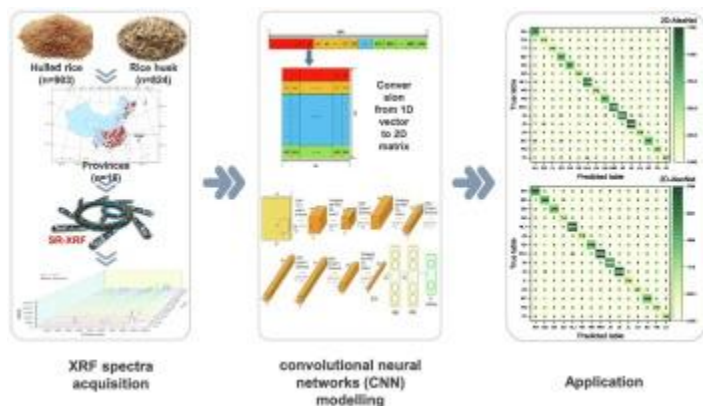
SR-XRF–deep learning framework offers a new versatile solution for food authentication.

Abstract

Accurate authentication of rice geographical origin is crucial for food safety and fraud prevention. Synchrotron radiation X-ray fluorescence (SR-XRF) spectroscopy was combined with deep learning to classify hulled rice ($n = 903$) and rice husks ($n = 824$) from 16 provinces in China. Distinct elemental fingerprints were observed, and PCA/t-SNE visualization confirmed clustering by origin. However, PCA-based discrimination was limited in resolving provinces with overlapping profiles, necessitating advanced nonlinear approaches. Three deep learning models—1D-CNN, 2D-VGG16, and 2D-AlexNet—were trained on SR-XRF spectra. 2D-AlexNet achieved the best results, with accuracies of 98.02% for hulled rice and 99.07% for husks, showing strong robustness across provinces.

Husk-based classification outperformed hulled rice, highlighting rice husks as a reliable sample matrix. The SR-XRF–deep learning framework provides a rapid, non-destructive, scalable tool for rice traceability, surpassing conventional chemometric or isotope-based methods.

Graphical abstract



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Introduction

Rice (*Oryza sativa* L.) is main staple food for half of population in the world, providing over 35% of caloric, 20% of protein and 3% of fat intake needs by a person, and also to facilitate their livelihoods(Basnet et al., 2016). In Asian countries such as China, Japan and Korea, there is more than 700 million tons of rice yields per annum, accounting for around 90% of rice output in global. China is the largest rice consumption countries more than quarter of the world's rice harvest(Basnet et al., 2016; Milovanovic & Smutka, 2017). Rice can be divided in to japonica (*Oryza sativa* L. *japonica*) and indica (*Oryza sativa* L. *indica*). In China, japonica rice is cultivated mainly in northeast plain and Yangtze River, and indica rice is cultivated mainly in southern China. Rice quality is based on the combined effects of inheritance of variety and environmental conditions(Feng, Li, Qin, Liao & Siddique, 2017). Rice grown in northeastern region in China has a good recognition and popularity for being of high quality and excellent taste, contributed to consumer preference for this rice producing areas. The provincial level represents an ideal scale for differentiation, as it encapsulates a macro-environment that shapes a distinctive elemental and biochemical fingerprint in the rice, making it a scientifically valid and practical standard for authentication. The higher quality of rice determines its higher market price, inducing the unfair competition behavior driven by huge profits and causing the adulteration and fraud

problem even serious food safety events for rice in China(Lam et al., 2013; Z. Liu et al., 2019).

Recent years, with the improvement of living standards, there is high consumer interest about the food safety and source authenticity of agricultural products such as rice(Zhi et al., 2023). Before 2023, China's traceability of agricultural product quality and safety just be confined to the corporation spontaneous behavior and government proposals. In a matter of practice, where there happens agricultural product safety problem accident, because of no strict quality traceability, it is often impossible to trace the specific origin of the defective products, so it has to destroy all the product in a huge area, and which cause large economic and reputation losses to the farmers who has standardized production. The new revised Agricultural Product Quality and Safety Law of the People's Republic of China (2022 Revision) establish the traceability cooperation mechanism for the quality and safety of agricultural product in order to strictly regulates food fraud and adulteration actions. Therefore, the geographical origin traceability of agricultural product is a necessary prerequisite for government decision-making and implementation(Guo et al., 2019).

In the field of food source analysis, the techniques applied in food traceability mainly include mineral elements analysis, spectroscopic technology, stable isotope techniques, metabonomics analysis, DNA technology, and other(Bai et al., 2023). Non-destructive spectroscopic techniques such as near-infrared (NIR), mid-infrared (MIR), and Raman spectroscopy are widely used for food authenticity and geographical origin traceability. These methods primarily detect molecular vibrations and functional groups associated

with organic components (e.g., C—H, O—H, N—H bonds in NIR/MIR; specific molecular fingerprints in Raman), offering advantages in portability, rapid analysis, minimal sample preparation, and low cost. They have been successfully applied to rice origin discrimination when combined with chemometrics or machine learning. However, their sensitivity to trace inorganic elements, particularly heavy metals and ultra-trace minerals at ppm or sub-ppm levels, is generally limited, as they rely on vibrational rather than elemental information, making it challenging to capture soil- and environment-derived inorganic fingerprints that are crucial for precise geographical provenance. In the rice production chain, the rice from different producer always present different elemental signatures, due to the characteristics of the surroundings (including soil, climate, seasonality, water et al.,) where it has been grown and mode of production which it has been used(M. Chen et al., 2022). The mineral elements content and chemometric model recognition have been used as a reliable and commonly used technology that has been used to ensure the authenticity of agricultural product provenance and detect food adulteration(Gonzalvez et al., 2011; Lange et al., 2019). In particular, multi-element

analysis method such as X-ray fluorescence (XRF)(T. Chen et al., 2021), inductively coupled plasma mass spectrometry (ICP-MS)(H. Zhao et al., 2017), and inductively coupled plasma optical emission spectrometry (ICP-OES)(Gonzalvez et al., 2011) has been extensively studied in the origin tracing of agricultural products such as rice, tea, wheat, and other agricultural products(H. Zhao et al., 2021). Among the multi-elements analysis technique, XRF is one of a non-destructive and solid-sampling analysis technique without chemical or digestion treatment, which is highly suitable for agricultural product traceability which need huge amounts of samples for statistics analysis. Its key strengths include non-destructive nature, broad elemental coverage (typically Na to U), and good sensitivity for many trace metals (often at ppm level) (Panebianco et al., 2022). Our previous works has reported the usage of commercial energy dispersive XRF (portable ED-XRF) to analyze the mineral elements in 296 soybean sample from adjacent geographic of northern China, and the correct rate of the identification model reached 96.2%(Lai et al., 2020). The commercial XRF has also been used for traceability and authentication for Basmati and Thai rice, and achieve a correct classification in the range from 94% to 100%(Dumitrascu et al., 2021). Therefore, XRF as particularly advantageous for rice origin authentication, where trace inorganic markers often provide more discriminative power than molecular signatures alone. While NIR/MIR/Raman excel in organic profiling and field applicability, XRF (especially SR-XRF) offers complementary strengths in inorganic multi-element profiling that are difficult to achieve with vibrational spectroscopies. However, conventional XRF systems are limited by moderate sensitivity and restricted elemental coverage, particularly for light elements. In contrast, synchrotron radiation XRF (SR-XRF) provides much higher brilliance, photo flux and a broad energy spectrum, enabling ultra-trace detection, reaching down to only a few atoms per square nanometer in nanoscale analyses (e.g., ~650 Ga atoms in doped silicon), whereas conventional XRF typically operates at the ppm level. Moreover, the wide energy range of SR-XRF allows simultaneous multi-element analysis, including lighter elements such as C, N, and O, which are generally

beyond the detection capability of conventional XRF that mainly covers Na to U elements(Vanhoof et al., 2025). In the recent years, synchrotron radiation -based techniques had been gradually used in agricultural field, offering rapid analysis, minimal sample requirements, and ultra-low detection limits(X. Li et al., 2021; Phedorin et al., 2000). Therefore, SR-XRF holds great potential for high-throughput, ultrasensitive agrometallomics researches(X. Li et al., 2021). Nevertheless, its application in the geographical origin of agricultural products remains very limited, leaving considerable room for further exploration and practical implementation.

In the field of agricultural product traceability and authentication, conventional chemometric techniques such as principal component analysis (PCA)(Lange et al., 2019),

linear discriminant analysis (LDA)(Sim et al., 2024), and partial least squares-discriminant analysis (PLS-DA)(S. Zhao et al., 2020) have long played a central role. These methods have shown value in classification tasks based on spectral data; however, they rely heavily on linear assumptions and manual feature extraction, which limit their ability to model complex, high-dimensional, and nonlinear datasets generated by advanced spectroscopic techniques such as near-infrared (NIR), hyperspectral imaging (HSI), and Raman spectroscopy(Jang et al., 2022; Pan et al., 2024; Q. Zhao et al., 2024). To address these limitations, machine learning (ML) methods have increasingly been explored in traceability modeling. Traditional ML algorithms, including support vector machines (SVM), random forests (RF), k-nearest neighbors (KNN) and other algorithms(Li et al., 2025), have demonstrated superior performance compared to conventional chemometrics in various agricultural origin identification tasks. These models can capture nonlinear relationships and often provide higher predictive accuracy and robustness. However, traditional ML approaches still rely on manual feature engineering and may struggle with large-scale datasets or highly complex spectral patterns(Maione, Barbosa Jr. & Barbosa, 2019). Deep learning (DL), a subset of ML, offers a more advanced and automated framework. DL models, such as multilayer perceptrons (MLPs) and convolutional neural networks (CNNs), can automatically learn hierarchical representations directly from raw input data without the need for handcrafted features(Ji et al., 2023; Lin et al., 2023). CNNs, in particular, require minimal prior knowledge or explicit feature design and are capable of extracting both local and global features from high-dimensional spectral inputs through an end-to-end training process(Chakravartula et al., 2022). In recent years, CNN-based architectures have shown great potential in the construction of traceability models. For instance, one-dimensional CNNs (1D-CNNs) are specifically designed for sequential data such as raw spectral vectors, effectively preserving and learning from their spectral continuity(Mu et al., 2021). A key advantage of 1D-CNNs lies in their ability to directly accept raw spectral data as input, offering a relatively straightforward and effective modeling approach(Hu et al., 2024). On the other hand, two-dimensional CNNs, such as AlexNet and VGGNet, can process transformed spectral data as images, allowing them to extract spatial and spectral features simultaneously(Hao et al., 2022). Motivated by the rapid advancement and maturity of image detection technologies compared to other domains, an emerging strategy involves converting spectral data into image-like formats. This transformation enables the application of powerful 2D deep learning models to leverage their superior capability in handling two-dimensional representations. A major reason for selecting 2D-CNNs is their ability to capture more comprehensive spectral information and accelerate processing speeds, which are critical for handling large-scale and complex spectral datasets(Zhou et al., 2022). These architectures provide complementary modeling

perspectives, offering a strong foundation for developing robust and scalable solutions for geographical origin authentication(Li et al., 2025).

A method named non-targeted metallomics (NTM) combined with machine learning algorithms (MLAs) was established for the screening cancer patients(He et al., 2022). Building on this concept, this study aims to develop and validate an effective strategy combining SRXRF and MLA for the traceability using 1797 rice samples (including 903 hulled rice and 894 rice husk) from 16 provinces of China. The present work makes several key contributions to the literature. It represents the first demonstration of integrating SR-XRF full spectra with deep learning, particularly by converting 1D spectra into 2D pseudo-images to leverage powerful image-recognition architectures such as AlexNet and VGG16, achieving high-accuracy classification across 16 Chinese provinces. Furthermore, by directly comparing hulled rice and rice husks as sample matrices, this study reveals that husk-based models attain slightly superior performance (99.07% vs. 98.02% accuracy), underscoring rice husks as a potentially more reliable and by-product-derived matrix for origin authentication. The work also achieves state-of-the-art performance (>98–99% accuracy) using raw SR-XRF spectra without preprocessing, fully exploiting the ultra-high sensitivity and multi-elemental detection capability of synchrotron radiation to capture subtle elemental fingerprints beyond conventional XRF or chemometric methods. Finally, it provides a rapid, versatile, and non-destructive SR-XRF–deep learning framework that offers clear advantages over traditional isotope or metabolomics-based approaches in throughput, cost-effectiveness, and applicability for large-scale food authentication and quality assurance systems. Once established, it would become an alternative tool for supporting government supervision for rice market and prevent mislabeling fraud of rice.

Section snippets

Sample collection

To ensure the reliability and stability, a total of 1797 rice samples (including 903 hulled rice samples and 894 rice husk sample) from 16 provinces were collected and which produced between 2020 and 2021, with the corresponding location. Fig. 1A shows the provinces located samples including in this study. These regions we selected represent the majority of rice production areas in China(Frolking et al., 2002). The provinces including Anhui (AH), Fujian (FJ), Guangdong (GD), Guangxi (GX),

Spectral characteristics of Rice samples

Synchrotron radiation X-ray fluorescence (SR-XRF) spectroscopy enables high-sensitivity, multi-elemental analysis. The raw spectra of hulled rice and rice husk samples spanned the full spectral range of 0–4096 channels, revealing clear peak structures corresponding

to characteristic elemental responses. We observed that the first 1024 channels contained more prominent and visually distinct spectral features. To better illustrate these patterns, 3D waterfall plots were generated based on the

Conclusion

This study confirms the feasibility of integrating SR-XRF spectroscopy with deep learning for accurate geographical traceability of rice in China. Both hulled rice and rice husk samples exhibited distinct elemental fingerprints that enabled reliable classification across multiple provinces. Dimensionality reduction analyses (PCA and t-SNE) revealed clear regional clustering, with husk samples showing more pronounced separation patterns. Among the tested models, the 2D-AlexNet architecture

CRedit authorship contribution statement

Xue Li: Writing – review & editing, Writing – original draft, Visualization, Formal analysis, Data curation, Conceptualization. **Chaojie Wei:** Writing – original draft, Visualization, Validation, Methodology. **Hongxin Xie:** Validation, Investigation, Data curation. **Haoxuan Luan:** Investigation. **Jinying Wang:** Investigation. **Yunbing Gao:** Visualization. **Yanan Zhao:** Visualization. **Wei Wang:** Writing – review & editing, Supervision, Conceptualization. **Yu-Feng Li:** Writing – review & editing, Supervision,

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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